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Lord

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(54) **ELECTRONIC VAPOR PROVISION DEVICE**

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CPC ***A24F 40/44*** (2020.01); ***A24F 40/40*** (2020.01); ***A24F 40/46*** (2020.01); *A24F 40/10* (2020.01)

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2,057,353	A	10/1936	Whittemore, Jr.
2,809,634	A	10/1957	Hirotada
2,937,648	A	5/1960	Meyer
2,991,788	A	7/1961	Brost
3,111,396	A	11/1963	Ball

(Continued)

FOREIGN PATENT DOCUMENTS

AT	508244	A4	12/2010
AU	6393173	A	6/1975

(Continued)

OTHER PUBLICATIONS

"Cambridge Dictionary Sleeve Definition", available at <dictionary/Cambridge.org/dictionary/English/sleeve>, Feb. 9, 2019.

(Continued)

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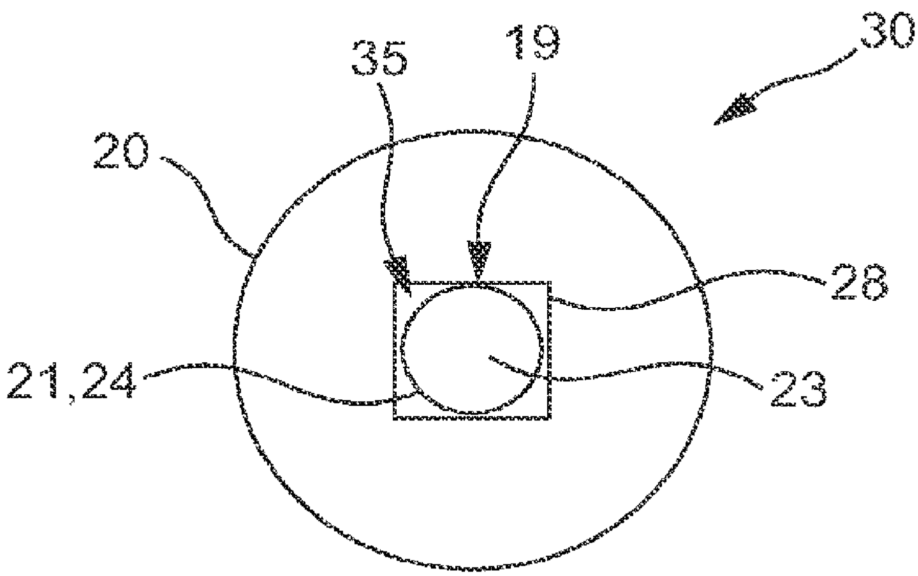
(57) **ABSTRACT**

An electronic vapor provision device comprising a power cell and a vaporizer, wherein the vaporizer comprises a heater and a heater support, wherein the heater is one the inside of the heater support.

9 Claims, 6 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

3,148,996	A	9/1964	Vukasovich et al.
3,239,117	A	3/1966	Letchworth
3,402,724	A	9/1968	Blount et al.
3,431,393	A	3/1969	Katsuda
3,433,632	A	3/1969	Elbert et al.
3,521,643	A	7/1970	Toth
3,604,428	A	9/1971	Moukaddem
3,804,100	A	4/1974	Fariello
3,844,199	A	10/1974	Block et al.
3,964,902	A	6/1976	Fletcher et al.
4,009,713	A	3/1977	Simmons et al.
4,031,906	A	6/1977	Knapp
4,094,119	A	6/1978	Sullivan
4,145,001	A	3/1979	Weyenberg et al.
4,161,283	A	7/1979	Hyman
4,193,513	A	3/1980	Bull
4,219,031	A	8/1980	Full et al.
4,503,851	A	3/1985	Braunroth
4,588,976	A	5/1986	Jaselli
4,676,237	A	6/1987	Wood et al.
4,735,217	A	4/1988	Gerth et al.
4,827,950	A	5/1989	Banerjee et al.
4,830,028	A	5/1989	Lawson et al.
4,846,199	A	7/1989	Rose
4,848,374	A	7/1989	Chard et al.
4,885,129	A	12/1989	Leonard et al.
4,917,301	A	4/1990	Munteanu
4,922,901	A	5/1990	Brooks et al.
4,924,886	A	5/1990	Litzinger
4,947,874	A	8/1990	Brooks et al.
4,947,875	A	8/1990	Brooks et al.
4,978,814	A	12/1990	Honour
5,027,837	A	7/1991	Clearman et al.
5,046,514	A	9/1991	Bolt
5,060,671	A	10/1991	Counts et al.
5,065,776	A	11/1991	Lawson et al.
5,095,647	A	3/1992	Zobele et al.
5,095,921	A	3/1992	Losee et al.
5,099,861	A	3/1992	Clearman et al.
5,115,823	A	5/1992	Keritsis
5,121,881	A	6/1992	Lembeck
5,129,409	A	7/1992	White et al.
5,144,962	A	9/1992	Counts et al.
5,167,242	A	12/1992	Turner et al.
5,179,966	A	1/1993	Losee et al.
5,247,947	A	9/1993	Clearman et al.
5,322,075	A	6/1994	Deevi et al.
5,369,723	A	11/1994	Counts et al.
5,388,574	A	2/1995	Ingebrethsen
5,390,864	A	2/1995	Alexander
5,415,186	A	5/1995	Casey et al.
5,415,486	A	5/1995	Wouters et al.
5,479,948	A	1/1996	Counts et al.
5,497,792	A	3/1996	Prasad et al.
5,501,236	A	3/1996	Hill et al.
5,505,214	A	4/1996	Collins et al.
5,540,241	A	7/1996	Kim
5,553,791	A	9/1996	Alexander
5,611,360	A	3/1997	Tang
5,636,787	A	6/1997	Gowhari
5,649,554	A	7/1997	Sprinkel et al.
5,666,977	A	9/1997	Higgins et al.
5,692,291	A	12/1997	Deevi et al.
5,692,526	A	12/1997	Adams et al.
5,743,251	A	4/1998	Howell et al.
5,865,185	A	2/1999	Collins et al.
5,954,060	A	9/1999	Cardarelli
6,095,505	A	8/2000	Miller
6,155,268	A	12/2000	Takeuchi
6,275,650	B1	8/2001	Lambert
6,280,793	B1	8/2001	Atwell et al.
6,532,965	B1	3/2003	Abhulimen et al.
6,652,804	B1	11/2003	Neumann et al.
6,681,998	B2	1/2004	Sharpe et al.
6,701,921	B2	3/2004	Sprinkel et al.
6,790,496	B1	9/2004	Levander et al.
7,100,618	B2	9/2006	Dominguez
7,112,712	B1	9/2006	Ancell
7,263,228	B2	8/2007	Mori
7,400,940	B2	7/2008	Mcrae et al.
7,540,286	B2	6/2009	Cross et al.
7,767,698	B2	8/2010	Warchol et al.
7,832,410	B2	11/2010	Hon
7,992,554	B2	8/2011	Radomski et al.
8,156,944	B2	4/2012	Han
8,205,622	B2	6/2012	Pan
8,365,742	B2	2/2013	Hon
8,375,957	B2	2/2013	Hon
8,393,331	B2	3/2013	Hon
8,430,106	B2	4/2013	Potter et al.
8,490,628	B2	7/2013	Hon
8,511,318	B2	8/2013	Hon
8,689,805	B2	4/2014	Hon
8,752,545	B2	6/2014	Buchberger
8,833,364	B2	9/2014	Buchberger
8,948,578	B2	2/2015	Buchberger
8,975,764	B1	3/2015	Abehasera
9,623,205	B2	4/2017	Buchberger
9,943,108	B2	4/2018	Lord
9,974,335	B2	5/2018	Lord
10,111,466	B2	10/2018	Lord
10,278,421	B2	5/2019	Lord
10,582,729	B2	3/2020	Lord
10,588,354	B2	3/2020	Lord
2001/0042546	A1	11/2001	Umeda et al.
2002/0016370	A1	2/2002	Shytle et al.
2002/0079309	A1	6/2002	Cox et al.
2003/0005620	A1	1/2003	Ananth et al.
2003/0049025	A1	3/2003	Neumann et al.
2003/0063902	A1	4/2003	Pedrotti et al.
2003/0079309	A1	5/2003	Vandenbelt et al.
2003/0106552	A1	6/2003	Sprinkel et al.
2003/0200964	A1	10/2003	Blakley et al.
2004/0031485	A1	2/2004	Rustad et al.
2004/0065749	A1	4/2004	Kotary et al.
2004/0129793	A1	7/2004	Nguyen et al.
2004/0151598	A1	8/2004	Young et al.
2005/0204799	A1	9/2005	Koch
2005/0268911	A1	12/2005	Cross et al.
2006/0078477	A1	4/2006	Althouse et al.
2006/0131439	A1	6/2006	Lakatos et al.
2007/0014549	A1	1/2007	Demarest et al.
2007/0062548	A1	3/2007	Horstmann et al.
2007/0102013	A1	5/2007	Adams et al.
2007/0107879	A1	5/2007	Radomski et al.
2007/0137667	A1	6/2007	Zhuang et al.
2007/0155255	A1	7/2007	Galauner et al.
2007/0283972	A1	12/2007	Monsees et al.
2008/0092912	A1	4/2008	Robinson et al.
2008/0216828	A1	9/2008	Wensley et al.
2008/0241255	A1	10/2008	Rose et al.
2009/0095311	A1	4/2009	Han
2009/0188490	A1	7/2009	Han
2009/0272379	A1	11/2009	Thorens et al.
2009/0288668	A1	11/2009	Inagaki
2009/0293888	A1	12/2009	Williams et al.
2009/0293892	A1	12/2009	Williams et al.
2009/0302019	A1	12/2009	Selenski et al.
2010/0006113	A1	1/2010	Urtsev et al.
2010/0024834	A1	2/2010	Oglesby et al.
2010/0059070	A1	3/2010	Potter et al.
2010/0065653	A1	3/2010	Wingo et al.
2010/0083959	A1	4/2010	Siller
2010/0108059	A1	5/2010	Axelsson et al.
2010/0236546	A1	9/2010	Yamada et al.
2011/0005535	A1	1/2011	Xiu
2011/0011396	A1	1/2011	Fang
2011/0036363	A1	2/2011	Urtsev et al.
2011/0094523	A1*	4/2011	Thorens H05B 1/0244 131/194
2011/0126848	A1	6/2011	Zuber et al.
2011/0155153	A1	6/2011	Thorens et al.
2011/0168194	A1	7/2011	Hon
2011/0209717	A1	9/2011	Han

(56) **References Cited**

U.S. PATENT DOCUMENTS

2011/0226236	A1	9/2011	Buchberger
2011/0232654	A1	9/2011	Mass
2011/0277756	A1	11/2011	Terry et al.
2011/0277757	A1	11/2011	Terry et al.
2011/0290267	A1	12/2011	Yamada et al.
2011/0297166	A1	12/2011	Takeuchi et al.
2011/0303231	A1	12/2011	Li et al.
2012/0006343	A1	1/2012	Renaud et al.
2012/0111347	A1	5/2012	Hon
2012/0145169	A1	6/2012	Wu
2012/0179512	A1	7/2012	Okeeffe
2012/0227753	A1	9/2012	Newton
2012/0234821	A1	9/2012	Shimizu
2012/0255567	A1	10/2012	Rose et al.
2012/0260927	A1	10/2012	Liu
2012/0279512	A1	11/2012	Hon
2012/0285475	A1	11/2012	Liu
2012/0285476	A1	11/2012	Hon
2013/0037041	A1	2/2013	Worm et al.
2013/0056013	A1	3/2013	Terry et al.
2013/0074857	A1	3/2013	Buchberger
2013/0081619	A1	4/2013	Seakins et al.
2013/0081623	A1	4/2013	Buchberger
2013/0192615	A1	8/2013	Tucker et al.
2013/0192623	A1	8/2013	Tucker et al.
2013/0213417	A1	8/2013	Chong et al.
2013/0213419	A1	8/2013	Tucker et al.
2013/0284192	A1	10/2013	Peleg et al.
2013/0298905	A1	11/2013	Levin et al.
2013/0306085	A1	11/2013	Sanchez et al.
2013/0333700	A1	12/2013	Buchberger
2013/0340779	A1	12/2013	Liu
2014/0000638	A1	1/2014	Sebastian et al.
2014/0007863	A1	1/2014	Chen
2014/0024834	A1	1/2014	Mergelsberg et al.
2014/0060528	A1	3/2014	Liu
2014/0060529	A1	3/2014	Zhang
2014/0060554	A1	3/2014	Collett et al.
2014/0060555	A1	3/2014	Chang et al.
2014/0069444	A1	3/2014	Cyphert et al.
2014/0202454	A1	7/2014	Buchberger
2014/0209105	A1	7/2014	Sears et al.
2014/0238396	A1	8/2014	Buchberger
2014/0238423	A1	8/2014	Tucker et al.
2014/0238424	A1	8/2014	Macko et al.
2014/0261490	A1	9/2014	Kane
2014/0270730	A1	9/2014	Depiano et al.
2014/0283825	A1	9/2014	Buchberger
2014/0286630	A1	9/2014	Buchberger
2014/0299125	A1	10/2014	Buchberger
2014/0299142	A1	10/2014	Dincer et al.
2014/0338680	A1	11/2014	Abramov et al.
2015/0020831	A1	1/2015	Weigensberg et al.
2015/0114411	A1	4/2015	Buchberger
2015/0150302	A1	6/2015	Metrangolo et al.
2015/0157055	A1	6/2015	Lord
2015/0196058	A1	7/2015	Lord
2015/0201675	A1	7/2015	Lord
2015/0208728	A1	7/2015	Lord
2015/0245654	A1	9/2015	Memari et al.
2015/0258288	A1	9/2015	Sullivan
2015/0333552	A1	11/2015	Alarcon
2015/0333561	A1	11/2015	Alarcon
2016/0073693	A1	3/2016	Reevell
2016/0106154	A1	4/2016	Lord
2016/0106155	A1	4/2016	Reevell
2016/0250201	A1	9/2016	Rose et al.
2016/0278436	A1	9/2016	Verleur et al.
2016/0295923	A1	10/2016	Lin
2016/0353804	A1	12/2016	Lord
2017/0042245	A1	2/2017	Buchberger et al.
2017/0114965	A1	4/2017	Maglica et al.
2017/0143042	A1	5/2017	Batista et al.
2017/0173278	A1	6/2017	Buchberger
2017/0197043	A1	7/2017	Buchberger
2017/0197044	A1	7/2017	Buchberger
2017/0197046	A1	7/2017	Buchberger
2017/0208865	A1	7/2017	Nettenstrom et al.
2017/0251725	A1	9/2017	Buchberger et al.
2018/0192705	A1	7/2018	Lord
2018/0199618	A1	7/2018	Fuisz et al.
2018/0235284	A1	8/2018	Lord
2021/0100285	A1	4/2021	Spencer et al.

FOREIGN PATENT DOCUMENTS

CA	2309376	A1		11/2000
CA	2864238	A1		8/2013
CA	103974639	A		8/2014
CH	698603	B1		9/2009
CN	1040496	A		3/1990
CN	2082939	U		8/1991
CN	2092880	U		1/1992
CN	2220168	Y		2/1996
CN	2249068	Y		3/1997
CN	1205849	A		1/1999
CN	2719043	Y		8/2005
CN	1925757	A		3/2007
CN	201054977	Y		5/2008
CN	201079011	Y		7/2008
CN	101277623	A		10/2008
CN	201238609	Y		5/2009
CN	101500443	A		8/2009
CN	201375023	Y		1/2010
CN	201379072	Y		1/2010
CN	201468000	U		5/2010
CN	101795505	A		8/2010
CN	101843368	A	*	9/2010
CN	101878958	A		11/2010
CN	202085723	U		12/2011
CN	202172846	U		3/2012
CN	102655773	A		9/2012
CN	202722498	U		2/2013
CN	202750708	U		2/2013
CN	103070472	A		5/2013
CN	203168033	U		9/2013
CN	103750573	A		4/2014
CN	103929988	A		7/2014
CN	103974369	A		8/2014
CN	104095293	A		10/2014
CN	203943069	U		11/2014
CN	204120237	U		1/2015
CN	104349687	A		2/2015
CN	106102863	A		11/2016
DE	822964	C		11/1951
DE	1950439	A1		4/1971
DE	3148335	A1		7/1983
DE	3218760	A1		12/1983
DE	3844022	C1		2/1990
DE	3936687	A1		5/1990
DE	29713866	U1		10/1997
DE	19630619	A1		2/1998
DE	19654945	A1		3/1998
DE	10330681	B3		6/2004
DE	202006013439	U1		10/2006
DE	102006004484	A1		8/2007
DE	102007011120	A1		9/2008
DE	202013100606	U1		2/2013
EA	015651	B1		10/2011
EA	201100197	A1		3/2012
EP	0280262	A2		8/1988
EP	0295122	A2		12/1988
EP	0358002	A2		3/1990
EP	0444553	A2		9/1991
EP	0488488	A1		6/1992
EP	0532194	A1		3/1993
EP	0712584	A2		5/1996
EP	0845220	A1		6/1998
EP	0893071	A1		1/1999
EP	1166814	A2		1/2002
EP	1166847	A2		1/2002
EP	1283062	A1		2/2003
EP	0845220	B1		9/2003
EP	1486226	A2		12/2004

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

EP	1736065	A1	12/2006
EP	2018886	A1	1/2009
EP	2022349	A1	2/2009
EP	1736065	B1	6/2009
EP	2113178	A1	11/2009
EP	2119375	A1	11/2009
EP	2327318	A1	6/2011
EP	2340729	A1	7/2011
EP	2394520	A1	12/2011
EP	2404515	A1	1/2012
EP	2444112	A1	4/2012
EP	2444411	A1	4/2012
EP	2695531	A1	2/2014
EP	2698070	A1	2/2014
EP	2762019	A1	8/2014
EP	2835062	A1	2/2015
EP	2939553	A1	11/2015
EP	2083643	B1	9/2017
FR	960469	A	4/1950
GB	25575		3/1912
GB	1313525	A	4/1973
GB	2333466	A	7/1999
GB	2488257	A	8/2012
GB	2496105	A	5/2013
HK	1196511	A1	12/2014
HK	1226611	A	10/2017
JP	S5130900	U	3/1976
JP	S5752456	A	3/1982
JP	S59106340	A	6/1984
JP	S6196763	A	5/1986
JP	S6196765	A	5/1986
JP	H02124081	A	5/1990
JP	H0339077	A	2/1991
JP	H05103836	A	4/1993
JP	H05309136	A	11/1993
JP	H06315366	A	11/1994
JP	H07502188	A	3/1995
JP	H0878142	A	3/1996
JP	H08299862	A	11/1996
JP	H1189551	A	4/1999
JP	H11507234	A	6/1999
JP	2002527153	A	8/2002
JP	3392138	B2	3/2003
JP	2004332069	A	11/2004
JP	2005537918	A	12/2005
JP	2006504431	A	2/2006
JP	2007259864	A	10/2007
JP	2007267749	A	10/2007
JP	2009502136	A	1/2009
JP	2009504431	A	2/2009
JP	3153675	U	9/2009
JP	2009537119	A	10/2009
JP	2010520742	A	6/2010
JP	3164992	U	12/2010
JP	2011518567	A	6/2011
JP	2012517229	A	8/2012
JP	5130900	B2	1/2013
JP	2013545473	A	12/2013
JP	2014076065	A	5/2014
JP	2014525237	A	9/2014
JP	2015506170	A	3/2015
JP	2015524257	A	8/2015
KR	20050037919	A	4/2005
KR	20110006928	A	1/2011
KR	20110006928	U	7/2011
KR	101081481	B1	11/2011
RU	2004116065	A	6/2005
RU	2311859	C2	12/2007
RU	2336001	C2	10/2008
RU	89927	U1	12/2009
RU	94815	U1	6/2010
RU	103281	U1	4/2011
RU	2420290	C2	6/2011
RU	110608	U1	11/2011
RU	115629	U1	5/2012
RU	122000	U1	11/2012
RU	124120	U1	1/2013
RU	2480485	C2	4/2013
RU	145715	U1	9/2014
RU	158129	U1	12/2015
SU	1641182	A3	4/1991
TW	201225862	A	7/2012
WO	9406313	A1	3/1994
WO	9502712	A2	1/1995
WO	9527412	A1	10/1995
WO	9632854	A2	10/1996
WO	9639880	A1	12/1996
WO	9748293	A1	12/1997
WO	9836651	A1	8/1998
WO	0009188	A1	2/2000
WO	0021598	A1	4/2000
WO	02058747	A1	8/2002
WO	2002061701	A1	8/2002
WO	03028409	A1	4/2003
WO	03050405	A1	6/2003
WO	03083283	A1	10/2003
WO	03101454	A1	12/2003
WO	2004022128	A2	3/2004
WO	2004022242	A1	3/2004
WO	2004022243	A1	3/2004
WO	2004080216	A1	9/2004
WO	2005106350	A2	11/2005
WO	2006048774	A1	5/2006
WO	2006082571	A1	8/2006
WO	2006124757	A2	11/2006
WO	2007012007	A2	1/2007
WO	2007042941	A2	4/2007
WO	2007078273	A1	7/2007
WO	2007131449	A1	11/2007
WO	2008015441	A1	2/2008
WO	2008029381	A2	3/2008
WO	2009015410	A1	2/2009
WO	2009022232	A2	2/2009
WO	2009132793	A1	11/2009
WO	2010045670	A1	4/2010
WO	2010045671	A1	4/2010
WO	2010091593	A1	8/2010
WO	2011060788	A1	5/2011
WO	2011079932	A1	7/2011
WO	2011106788	A2	9/2011
WO	2011107737	A1	9/2011
WO	2011109849	A1	9/2011
WO	2011124033	A1	10/2011
WO	2011137453	A2	11/2011
WO	2011146372	A2	11/2011
WO	2011160788	A1	12/2011
WO	2012025496	A1	3/2012
WO	2012072264	A1	6/2012
WO	2012072762	A1	6/2012
WO	2012156700	A1	11/2012
WO	2013034453	A1	3/2013
WO	2013034460	A1	3/2013
WO	2013057185	A1	4/2013
WO	2013060784	A2	5/2013
WO	2013076098	A2	5/2013
WO	2013082173	A1	6/2013
WO	2013083631	A1	6/2013
WO	2013083634	A1	6/2013
WO	2013098395	A1	7/2013
WO	2013116558	A1	8/2013
WO	2013116571	A1	8/2013
WO	2013148810	A1	10/2013
WO	2013149404	A1	10/2013
WO	2013178766	A1	12/2013
WO	2014061477	A1	4/2014
WO	2014104078	A1	7/2014
WO	2014106093	A1	7/2014
WO	2014130695	A1	8/2014
WO	2014136872	A1	9/2014
WO	2014140320	A1	9/2014
WO	2014150131	A1	9/2014
WO	2015117702	A1	8/2015

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

WO	2016156493	A2	10/2016
WO	2016162446	A1	10/2016
WO	2017055866	A1	4/2017

OTHER PUBLICATIONS

"Canada Office Action, Application No. 2,878,973, mailed Jan. 22, 2016".
"Canadian Office Action, Application No. 2,878,959, dated Jan. 18, 2016".
"Chinese First Office Action for Chinese Application No. 200980152395.4 date issued Dec. 3, 2012".
"Chinese Office Action and Chinese Search Report, Chinese Application No. CN201680020844.X, mailed Jun. 24, 2019".
"Chinese Office Action and Search Report, Chinese Application No. 201680020842.0, mailed Jun. 21, 2019".
"Chinese Office Action, Application No. 201680020844.X, dated Jul. 2, 2019".
"Chinese Office Action, Chinese Application No. 201380038055.5, dated Jul. 11, 2017".
"Chinese Office Action, Chinese Application No. 201380038055.5, mailed on Apr. 18, 2016".
"Chinese Office Action, Chinese Application No. 201680020758.9, mailed Jul. 23, 2019".
"Chinese Second Office Action and Search Report, Application No. 201680020844.X, dated May 22, 2020".
"Communication pursuant to Article 94(3) EPC for European Application No. 16189742.6, mailed on Dec. 4, 2020".
"Decision to Grant in Russian Application No. 2014120213, dated Oct. 26, 2016".
"Decision to Grant mailed Oct. 24, 2019 for Russian Application No. 2019118770".
"European Extended Search Report, Application No. 18195423.1, dated Jan. 29, 2019".
"European Notice of Opposition, Application No. EP2871984, dated Jun. 6, 2017".
"European Search Report for European Application No. 16189742.6 mailed Mar. 17, 2017".
"European Search Report received for European Patent Application No. 16166656.5 dated Oct. 11, 2016".
"Exam Report from European Application 16189742.6-1006, dated Dec. 19, 2019".
"Examination Report for Australian Application No. 2015293686, dated Jul. 25, 2018".
"Examination Report for European Application No. 15741289.1, dated Jun. 15, 2018".
"Extended European Search Report for Application No. 16177005.2, mailed on Oct. 26, 2016".
"Extended European Search Report for European Application No. 15178588.8, mailed on Apr. 22, 2016".
"Extended European Search Report received for European Patent Application No. 17189951.1, mailed on Jan. 4, 2018".
"Extended European Search Report, European Application No. 19174777.3, mailed Nov. 11, 2019".
"GB Search Report, Application No. GB1505593.2, mailed Sep. 22, 2015,".
"GB Search Report, GB Application No. GB1505595.7, mailed Oct. 20, 2015".
"International Preliminary Report on Patentability for Appl. No. PCT/EP2016/057064, mailed on Oct. 12, 2017".
"International Preliminary Report on Patentability for Application No. PCT/AT2012/000017, issued on Aug. 13, 2013".
"International Preliminary Report on Patentability for Application No. PCT/EP2012/070647, mailed on May 1, 2014".
"International Preliminary Report on Patentability for Application No. PCT/EP2016/057060, mailed on Jul. 12, 2017".
"International Preliminary Report on Patentability for Application No. PCT/EP2016/057097, mailed on Oct. 12, 2017".
"International Preliminary Report on Patentability for Application No. PCT/GB2014/051332, mailed on Nov. 12, 2015".
"International Preliminary Report on Patentability for Application No. PCT/GB2014/051333, completed on Aug. 5, 2015".
"International Preliminary Report on Patentability for Application No. PCT/GB2014/051334, mailed on Nov. 12, 2015".
"International Preliminary Report on Patentability for corresponding International Application No. PCT/GB2015/051213 mailed on Jul. 14, 2016".
"International Preliminary Report on Patentability, mailed Oct. 27, 2014, for Application No. PCT/EP2013/064952, filed Jul. 15, 2013".
"International Preliminary Report on Patentability, mailed Oct. 31, 2014, for Application No. PCT/EP2013/064950, filed Jul. 15, 2013.".
"International Search Report and Written Opinion for Application No. PCT/EP2016/057097, mailed on Sep. 28, 2016".
"International Search Report and Written Opinion received for PCT Application No. PCT/EP2012/003103, mailed on Nov. 26, 2012".
"International Search Report and Written Opinion received for PCT Application No. PCT/GB2014/051333, mailed on Jul. 17, 2014".
"International Search Report and Written Opinion received for PCT Application No. PCT/GB2014/051334 mailed Jul. 21, 2014".
"International Search Report and Written Opinion received for PCT Patent Application No. PCT/GB2014/051332, mailed on Jul. 21, 2014".
"International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2012/070647, mailed on Feb. 6, 2013".
"International Search Report and Written Opinion received for PCT Patent Application No. PCT/AT2012/000017, mailed on Jul. 3, 2012".
"International Search Report and Written Opinion, mailed Dec. 2, 2013, for Application No. PCT/EP2013/064950, filed Jul. 15, 2013".
"International Search Report and Written Opinion, mailed Oct. 11, 2013, for Application No. PCT/EP2013/064952, filed Jul. 15, 2013".
"International Search Report and Written Opinion, PCT Application No. PCT/EP2016/057064, mailed Oct. 19, 2016".
"International Search Report for corresponding International Application No. PCT/GB2015/051213 mailed on Jul. 16, 2015".
"International Search Report received for PCT Patent Application No. PCT/AT2009/000413 mailed on Jan. 25, 2010".
"International Search Report received for PCT Patent Application No. PCT/AT2009/000414 mailed on Jan. 26, 2010".
"International Search Report, International Application No. PCT/EP2016/057060, mailed Sep. 28, 2016".
"Japanese Office Action for Japanese Application No. 2015-522066 dated Dec. 8, 2015".
"Japanese Office Action, Application No. 2015-522066, dated Jan. 5, 2016".
"Japanese Office Action, Application No. 2015-522064, dated Jan. 5, 2015".
"Japanese Office Action, Application No. 2017-551218, dated Oct. 30, 2018".
"Japanese Office Action, Application No. 2019-124231, mailed Oct. 27, 2020".
"Japanese Office Action, Japanese Application No. 2015-522064, mailed Dec. 28, 2015".
"Japanese Office Action, Japanese Application No. 2017-551205, mailed Oct. 2, 2018".
"Japanese Office Action, Japanese Application No. 2017-551206, mailed Oct. 23, 2019".
"Japanese Office Action, Japanese Application No. 2017551218, mailed Aug. 6, 2019".
"Japanese Office Action, Japanese Application No. 2015-522065, dated Jan. 5, 2016".
"Korean Decision for Refusal for Korean Application No. KR2020110006928 dated Jan. 10, 2019".
"Korean Notice Trial Decision to Reject for Korean Application No. 10-2015-7001257 mailed Aug. 14, 2019".
"Korean Office Action, Application No. 10-2015-7001256, dated Sep. 7, 2016".
"Korean Office Action, Application No. 10-2015-7001257, dated Sep. 8, 2016".

(56) **References Cited**

OTHER PUBLICATIONS

"Korean Office Action, Application No. 10-2017-7034160, dated Jul. 18, 2018".
"Notice of Opposition Letter from EPO Opposition against for EP2358418 mailed on Mar. 1, 2017".
"Office Action dated Aug. 24, 2018 for Chinese Application No. 201580040255.3".
"Office Action for Canadian Application No. 2,954,848, dated Dec. 18, 2017".
"Office Action for Chinese Application No. 201811153475.9, mailed on Apr. 22, 2021".
"Office Action for Japanese Application No. 2017-504040, dated Feb. 22, 2018".
"Office Action for Japanese Application No. 2017-504040, dated Oct. 9, 2018".
"Office Action for Japanese Application No. 2018-206299, dated Apr. 13, 2021".
"Office Action for Japanese Application No. 2018-206299, dated Oct. 6, 2020".
"Office Action mailed Mar. 15, 2018 for Korean Application No. 2017-7002235".
"Office Action received for Chinese Patent Application No. 201480024988.3, mailed on Dec. 30, 2016".
"Office Action received for Chinese Patent Application No. 201480024978.X mailed on Jan. 18, 2017".
"Office Action received for Chinese Patent Application No. 201480024988.3, mailed on Sep. 11, 2017".
"Office Action received for Russian Patent Application No. 2015146847, mailed on Sep. 22, 2017".
"Reasons for Rejection received for Japanese Patent Application No. 2015-137361, mailed on May 31, 2016".
"Reasons for Rejection received for Japanese Patent Application No. 2011-532464, mailed on Oct. 7, 2013".
"Reasons for Rejection received for Japanese Patent Application No. 2014-179732, mailed on Sep. 8, 2015".
"Reasons for Rejection received for Japanese Patent Application No. 2016-134648 mailed May 23, 2017".
"Russian Decision to Grant for Russian Application No. 2015100878 dated Sep. 19, 2016".
"Russian Decision to Grant, Application No. 2015100881/12, dated Apr. 6, 2016".
"Russian Office Action, Russian Application No. 2018118998, mailed Sep. 24, 2018".
"Search Report mailed May 17, 2020 for Chinese Application No. 201680020844.X".
"Search Report mailed Jan. 15, 2018, for Japanese Application No. 2017-504040".
"Search Report received for Russian Patent Application No. 2015146843/12 (072088), mailed on Apr. 24, 2017".
"Search Report, GB Application No. GB1505597.3, mailed Oct. 7, 2015".
"Translation of Chinese Second Office Action for Chinese Application No. 200980152395.4 date issued Aug. 20, 2013".
"Written Opinion for Application No. PCT/EP2016/057060, mailed on Sep. 28, 2016".
"Written Opinion of the International Preliminary Examining Authority for International Application No. PCT/GB2015/051213 mailed on Mar. 29, 2016".
"Written Opinion, International Application No. PCT/EP2016/057060, mailed Apr. 7, 2017".
Aerosols , "Pulmonary Pharmacology: Delivery Devices and Medications", available at http://www.ceu.org/cecourses/z98107/ch4.htm , Sep. 6, 2017 , 2 pages.
Christopher, Lord , "Application and File History for U.S. Appl. No. 14/415,552, filed Jan. 16, 2015".
Christopher, Lord , "Application and File History for U.S. Appl. No. 15/959,687, filed Apr. 23, 2018".
Diener Electronic , "Plasma Technology", The Company Diener Electronic GmbH+Co. KG, www.plasma.de , Oct. 17, 2017 , 19 Pages.
Dunn , et al. , "Heat Pipes", 4th edition, ISBN 0080419038 , 1994 , 14 Pages.
Kynol , "Standard Specifications of Kynol Activated Carbon Fiber Products", published by Kynol , Sep. 19, 2013 , 2 pages.
Lord , et al. , "Application and File History for U.S. Appl. No. 14/415,540, filed Jan. 16, 2015".
Lord , et al. , "Application and File History for U.S. Appl. No. 15/914,139, filed Mar. 7, 2018".
Rudolph, G , "The Influence of CO2 on the Sensory Characteristics of the Favor-System", BAT Cigarettenfabriken GmbH, http://legacy.library.ucsf.edu/tid/sla51f00 , 24 pages.
Sutton , "Application and File History for U.S. Appl. No. 15/563,065, filed Sep. 29, 2017".
Sutton , et al. , "Application and File History for U.S. Appl. No. 15/563,078, filed Sep. 29, 2017".
Sutton , et al. , "Application and File History for U.S. Appl. No. 15/563,086, filed Sep. 29, 2017".
European Notice Of Opposition in European Application No. EP18195423.1, dated Feb. 28, 2024, 37 pages.
European Search Report and Written Opinion in International Appln. No. 23176770.8, dated Nov. 16, 2023, 28 pages.
Office Action For Russian Application No. 2020124363, mailed on Feb. 17, 2021, 3 pages.

* cited by examiner

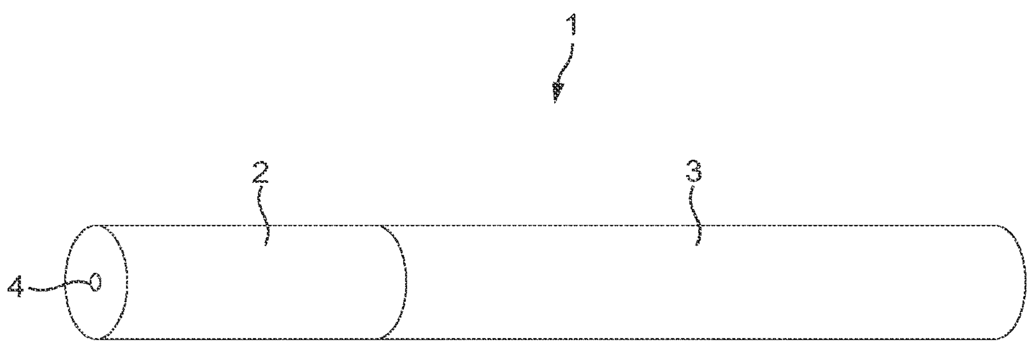


FIG. 1

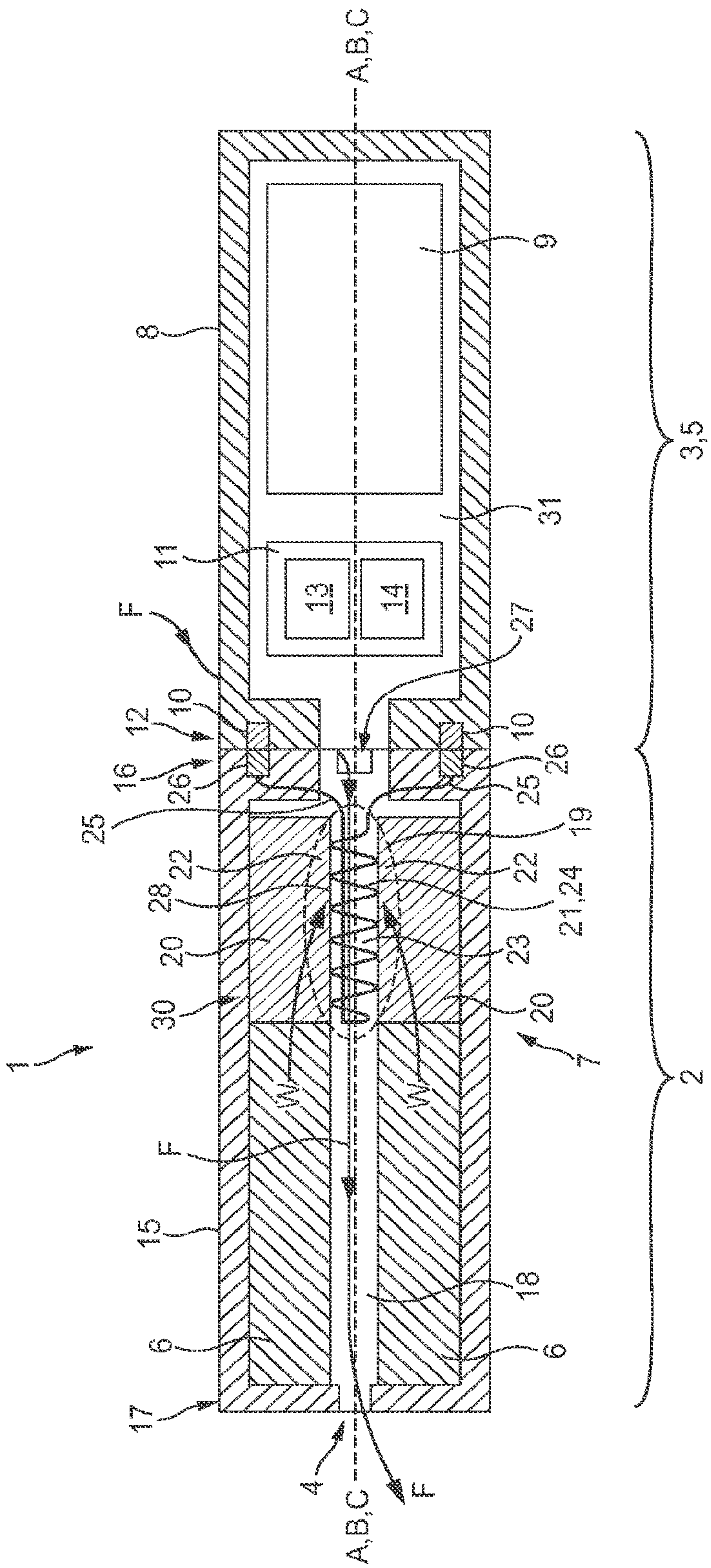

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FIG. 2

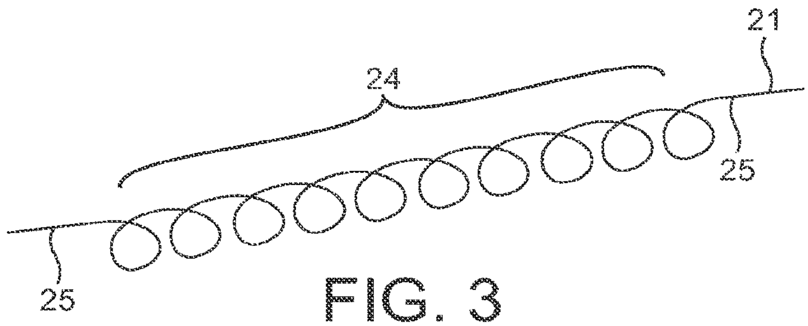
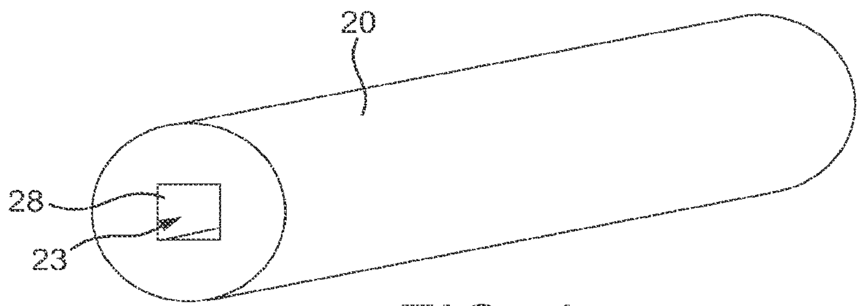


FIG. 3



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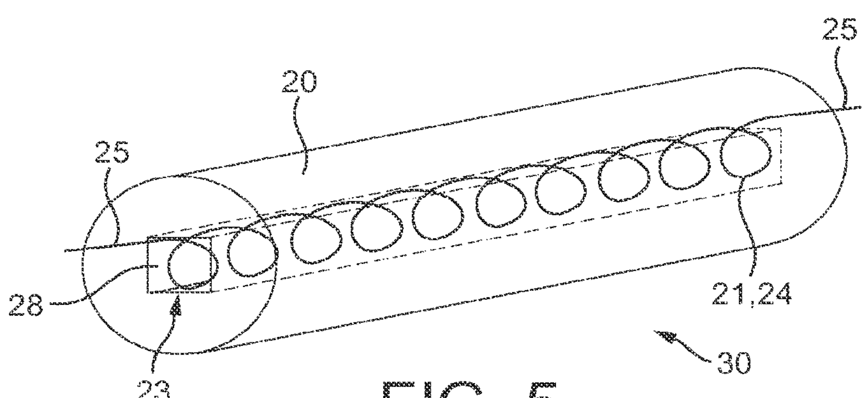


FIG. 5

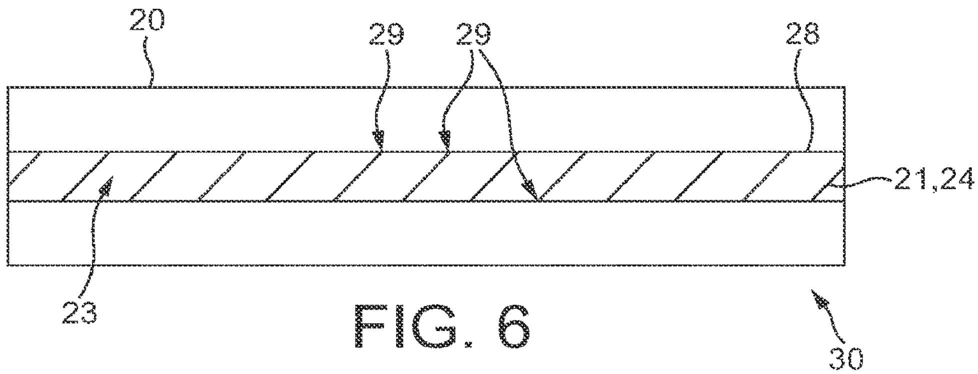
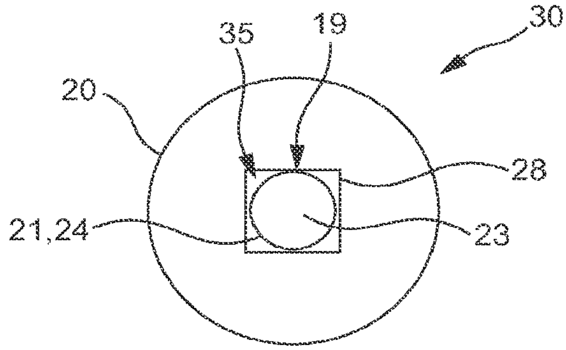


FIG. 6



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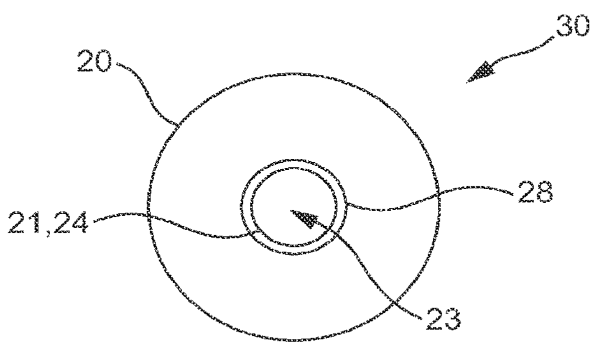

FIG. 8

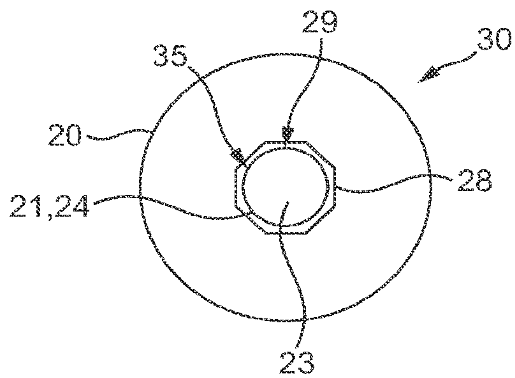

FIG. 9

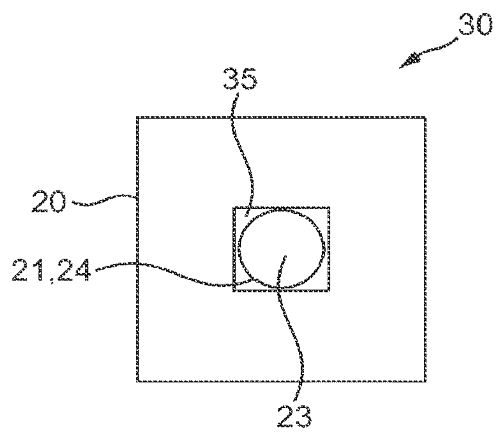

FIG. 10

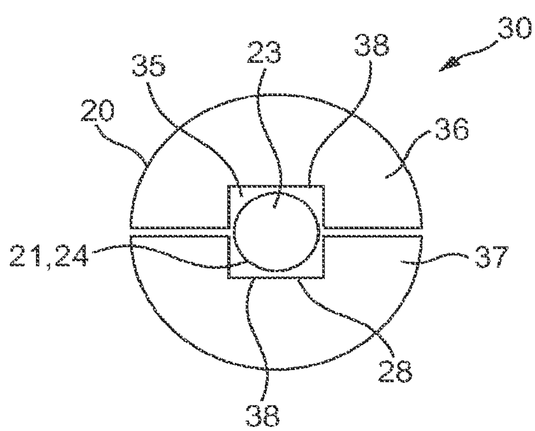

FIG. 11

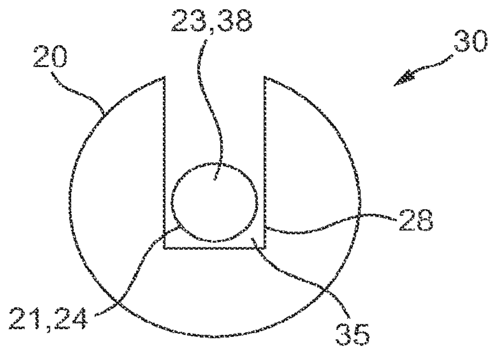

FIG. 12

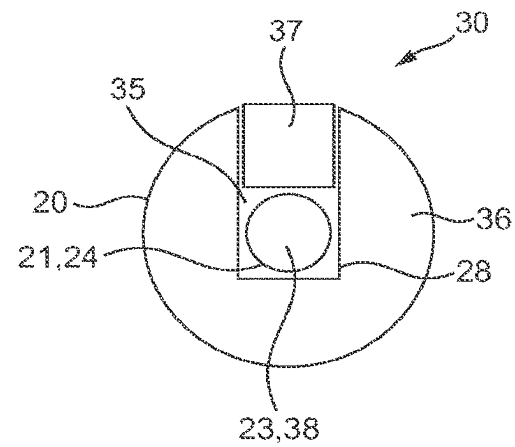

FIG. 13

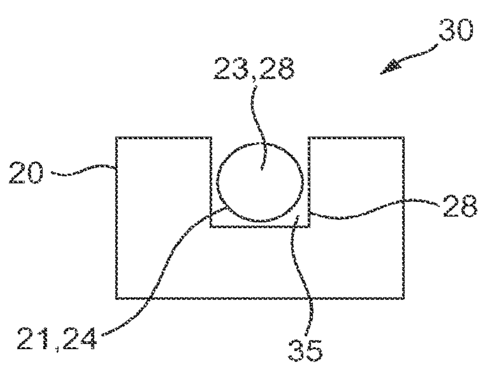

FIG. 14

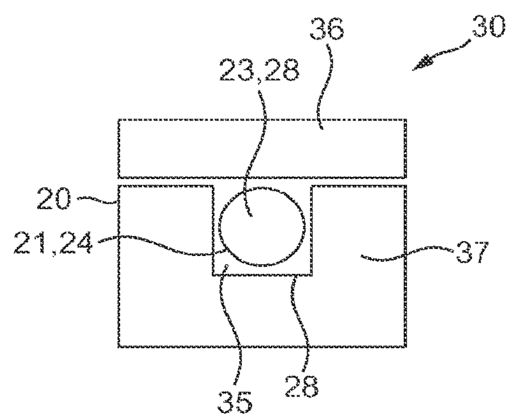

FIG. 15

ELECTRONIC VAPOR PROVISION DEVICE

RELATED APPLICATION

This application is a continuation of application Ser. No. 16/750,077, filed Jan. 23, 2020, which is a continuation of application Ser. No. 15/959,687, filed Apr. 23, 2018, now patented as U.S. Pat. No. 10,582,729, issued Mar. 10, 2022, which in turn is a divisional of application Ser. No. 14/415,540, filed Jan. 16, 2015, now patented as U.S. Pat. No. 9,974,335, issued May 22, 2018, which in turn is a National Phase entry of PCT Application No. PCT/EP2013/064950, filed Jul. 15, 2013, which claims the benefit of United Kingdom Application No. GB1212603.3, filed Jul. 16, 2012, each of which is hereby fully incorporated herein by reference.

TECHNICAL FIELD

The specification relates to electronic vapor provision devices.

BACKGROUND

Electronic vapor provision devices, such as electronic cigarettes, are typically cigarette-sized and typically function by allowing a user to inhale a nicotine vapor from a liquid store by applying a suction force to a mouthpiece. Some electronic vapor provision devices have an airflow sensor that activates when a user applies the suction force and causes a heater coil to heat up and vaporize the liquid.

SUMMARY

In an embodiment there is provided an electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support. One or more gaps may be provided between the heating element and the heating element support. Moreover, the electronic vapor provision device may have a mouthpiece section and the vaporizer may be part of the mouthpiece section. The heating element support may substantially fill the mouthpiece section.

In another embodiment there is provided a vaporizer for use in the vapor provision device, that comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support.

In another embodiment there is provided an electronic vapor provision device comprising a liquid store; a wicking element configured to wick liquid from the liquid store to a heating element for vaporizing the liquid; an air outlet for vaporized liquid from the heating element; and a heating element support, wherein the heating element is on the inside of the heating element support. The heating element support may be the wicking element. Moreover, the electronic vapor provision device may include a power cell for powering the heating element.

BRIEF DESCRIPTION OF THE DRAWINGS

For a better understanding of the disclosure, and to show how example embodiments may be carried into effect, reference will now be made to the accompanying drawings in which:

FIG. **1** is a side perspective view of an electronic cigarette.

FIG. **2** is a schematic sectional view of an electronic cigarette having a parallel coil.

FIG. **3** is a side perspective view of a heating element coil.

FIG. **4** is a side perspective view of an outer heating element support.

FIG. **5** is a side perspective view of a heating element coil within an outer heating element support.

FIG. **6** is a side sectional view of a heating element coil within an outer heating element support.

FIG. **7** is an end view of a heating element coil within an outer heating element support, where a central channel has a square cross-section.

FIG. **8** is an end view of a heating element coil within an outer heating element support, where a central channel has a circular cross-section.

FIG. **9** is an end view of a heating element coil within an outer heating element support, where a central channel has an octagonal cross-section.

FIG. **10** is an end view of a heating element coil within an outer heating element support having an outer square cross-section, where a central channel has a square cross-section.

FIG. **11** is an end view of a heating element coil within an outer heating support having two sections.

FIG. **12** is an end view of a heating element coil within a side channel of a heating element support.

FIG. **13** is an end view of a heating element coil within a side channel of a heating element support, with a second support section.

FIG. **14** is an end view of a heating element coil within a side channel of a heating element support having a rectangular cross-section.

FIG. **15** is an end view of a heating element coil within a side channel of a heating element support having a rectangular cross-section, with a second support section.

DETAILED DESCRIPTION

In an embodiment there is provided an electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support. The electronic vapor provision device may be an electronic cigarette.

Having a separate heating element and support allows a finer heating element to be constructed. This is advantageous because a finer heating element can be more efficiently heated. Having the heating element on the inside of the support means that a much smaller and narrower heating element can be used since space is not needed inside the heating element to house a support. This enables a much larger and therefore stronger support to be used.

The heating element may not be supported on its inside. Having a heating element that is not supported on its inside means that a support does not interfere with the heating element on its inner region. This provides a greater heating element surface area which thereby increases the vaporization efficiency.

The heating element support may be a liquid store. A combined support and liquid store has the advantage that liquid can be easily transferred from the liquid store to the heating element supported by the liquid store. Also, by eliminating the need for a separate support, the device can be made smaller or a larger liquid store can be utilized for increased capacity.

One or more gaps may be provided between the heating element and the heating element support. Providing a gap between the heating element and the heating element support allows liquid to gather, and thereby be stored, in the gap region for vaporization. The gap can also act to wick liquid onto the heating element. Also, providing a gap between the heating element and support means that a greater surface area of the heating element is exposed thereby giving a greater surface area for heating and vaporization.

The heating element may be in contact with the heating element support at two or more locations. Moreover, the heating element may be in contact with the heating element support at points along the length of the support.

The heating element support may be a rigid and/or a solid support. Furthermore, the heating element support may be porous. For example, the heating element support may be formed of porous ceramic material.

The heating element support may be elongated in a lengthwise direction. Moreover, the heating element support may have a support channel and the heating element may be located in the support channel. Furthermore, the support channel may run in a lengthwise direction of the heating element support.

The support channel may be an internal support channel. Moreover, the support channel may be a central support channel. Alternatively, the support channel may be a side support channel, located on a side of the heating element support.

The support channel may be substantially cylindrical. Moreover, the cross-sectional shape of the support channel may be circular. Alternatively, the cross-sectional shape of the support channel may be a polygon. Furthermore, the cross-sectional shape of the support channel may have 4 sides, 6 sides or 8 sides. Cross-sections are sections perpendicular to the elongated lengthwise direction. These various shapes of support channel provide natural gaps between the support and a heating element coil within the support channel. These gaps lead to increased wicking, liquid storage and vaporization.

The heating element support may comprise a first support section and a second support section. Moreover, the heating element may be supported by the first support section and the second support section. For example, the heating element may be supported between the first support section and the second support section. Furthermore, the support channel may be provided between the first support section and the second support section and the heating element may be in the support channel. The first support section may provide a first side of the support channel and the second support section may provide a second side of the support channel.

Providing a support that comprises two separate sections provides an easier method of assembly. It also enables a more accurate and consistent positioning of the heating element relative to the support.

The heating element may run along the length of the support channel. Moreover, the heating element may be in contact with the support channel at points along the length of the support channel. The heating element may be in contact with the surface of the support channel along the length of the support channel.

The heating element may be a heating coil, such as a wire coil. The heating coil may be coiled so as to be supported along its length by the heating element support. The turns of the heating coil may be supported by the heating element support. The turns of the heating coil may be in contact with the heating element support. A gap may be provided between the heating coil and the heating element support. Moreover, the gap may be between a coil turn and heating element support. Furthermore, gaps may be between coil turns and the heating element support.

By providing a gap between a coil turn and the support, liquid can be wicked into the gap and held in the gap for vaporization. In particular, liquid can be wicked by the spaces between coil turns and into the gap between a coil turn and the support.

The vaporizer may have a vaporization cavity configured such that in use the vaporization cavity is a negative pressure region. At least part of the heating element may be inside the vaporization cavity, or the heating element may be entirely inside the vaporization cavity. For example, the vaporization cavity may be inside the heating element support. Moreover, the vaporization cavity may be inside a channel of the heating element support. At least part of the vaporization cavity may be inside the heating element.

By having the heating element in the vaporization cavity, which in turn is a negative pressure region when a user inhales through the electronic vapor provision device, the liquid is directly vaporized and inhaled by the user.

The electronic vapor provision device may further include a mouthpiece section and the vaporizer may be part of the mouthpiece section. Moreover, the heating element support may substantially fill the mouthpiece section.

The liquid store may not comprise an outer liquid store container.

Since the support is on the outside of the coil and can act as a liquid store, a liquid store container is not needed in addition to the liquid store, and the heating element support can fill the mouthpiece section to give greater storage capacity and a more efficient device.

The electronic vapor provision device may further include a heating element connecting wire and the heating element support may include a heating element connecting wire support section.

The heating element support may be substantially cylindrical. The outer cross-sectional shape of the heating element support may be a circle. Alternatively, the outer cross-sectional shape of the heating element support may be a polygon. The outer cross-sectional shape of the heating element support may have 4 sides.

Referring to FIG. **1** there is shown an embodiment of the electronic vapor provision device **1** in the form of an electronic cigarette **1** comprising a mouthpiece **2** and a body **3**. The electronic cigarette **1** is shaped like a conventional cigarette having a cylindrical shape. The mouthpiece **2** has an air outlet **4** and the electronic cigarette **1** is operated when a user places the mouthpiece **2** of the electronic cigarette **1** in their mouth and inhales, drawing air through the air outlet **4**. Both the mouthpiece **2** and body **3** are cylindrical and are configured to connect to each other coaxially so as to form the conventional cigarette shape.

FIG. **2** shows an example of the electronic cigarette **1** of FIG. **1**. The body **3** is referred to herein as a battery assembly **5**, and the mouthpiece **2** includes a liquid store **6** and a vaporizer **7**. The electronic cigarette **1** is shown in its assembled state, wherein the detachable parts **2**, **5** are connected. Liquid wicks from the liquid store **6** to the vaporizer **7**. The battery assembly **5** provides electrical power to the vaporizer **7** via mutual electrical contacts of the battery assembly **5** and the mouthpiece **2**. The vaporizer **7** vaporizes the wicked liquid and the vapor passes out of the air outlet **4**. The liquid may for example comprise a nicotine solution.

The battery assembly **5** comprises a battery assembly casing **8**, a power cell **9**, electrical contacts **10** and a control circuit **11**.

The battery assembly casing **8** comprises a hollow cylinder which is open at a first end **12**. For example, the battery assembly casing **8** may be plastic. The electrical contacts **10** are located at the first end **12** of the casing **8**, and the power cell **9** and control circuit **11** are located within the hollow of the casing **8**. The power cell **9** may for example be a Lithium Cell.

The control circuit **11** includes an air pressure sensor **13** and a controller **14** and is powered by the power cell **9**. The controller **14** is configured to interface with the air pressure sensor **13** and to control provision of electrical power from the power cell **9** to the vaporizer **7**, via the electrical contacts **10**.

The mouthpiece **2** further includes a mouthpiece casing **15** and electrical contacts **26**. The mouthpiece casing **15** comprises a hollow cylinder which is open at a first end **16**, with the air outlet **4** comprising a hole in the second end **17** of the casing **15**. The mouthpiece casing **15** also comprises an air inlet **27**, comprising a hole near the first end **16** of the casing **15**. For example, the mouthpiece casing may be formed of aluminum.

The electrical contacts **26** are located at the first end of the casing **15**. Moreover, the first end **16** of the mouthpiece casing **15** is releasably connected to the first end **12** of the battery assembly casing **8**, such that the electrical contacts **26** of the mouthpiece **2** are electrically connected to the electrical contacts **10** of the battery assembly **5**. For example, the device **1** may be configured such that the mouthpiece casing **15** connects to the battery assembly casing **8** by a threaded connection.

The liquid store **6** is situated within the hollow mouthpiece casing **15** towards the second end **17** of the casing **15**. The liquid store **6** comprises a cylindrical tube of porous material saturated in liquid. The outer circumference of the liquid store **6** matches the inner circumference of the mouthpiece casing **15**. The hollow of the liquid store **6** provides an air passageway **18**. For example, the porous material of the liquid store **6** may comprise foam, wherein the foam is substantially saturated in the liquid intended for vaporization.

The vaporizer **7** comprises a vaporization cavity **19**, a heating element support **20** and a heating element **21**.

The vaporization cavity **19** comprises a region within the hollow of the mouthpiece casing **15** in which liquid is vaporized. The heating element **21** and a portion **22** of the support **20** are situated within the vaporization cavity **19**.

The heating element support **20** is configured to support the heating element **21** and to facilitate vaporization of liquid by the heating element **21**. The heating element support **20** is an outer support and is illustrated in FIGS. **4** to **7**. The support **20** comprises a hollow cylinder of rigid, porous material and is situated within the mouthpiece casing **15**, towards the first end **16** of the casing **15**, such that it abuts the liquid store **6**. The outer circumference of the support **20** matches the inner circumference of the mouthpiece casing **15**. The hollow of the support comprises a longitudinal, central channel **23** through the length of the support **20**. The channel **23** has a square cross-sectional shape, the cross-section being perpendicular to the longitudinal axis of the support.

The support **20** acts as a wicking element, as it is configured to wick liquid in the direction W from the liquid store **6** of the mouthpiece **2** to the heating element **21**. For example, the porous material of the support **20** may be nickel foam, wherein the porosity of the foam is such that the described wicking occurs. Once liquid wicks W from the liquid store **6** to the support **20**, it is stored in the porous material of the support **20**. Thus, the support **20** is an extension of the liquid store **6**.

The heating element **21** is formed of a single wire and comprises a heating element coil **24** and two leads **25**, as is illustrated in FIGS. **3**, **5**, **6** and **7**. For example, the heating element **21** may be formed of Nichrome. The coil **24** comprises a section of the wire where the wire is formed into a helix about an axis A. At either end of the coil **24**, the wire departs from its helical form to provide the leads **25**. The leads **25** are connected to the electrical contacts **26** and are thereby configured to route electrical power, provided by the power cell **9**, to the coil **24**.

The wire of the coil **24** is approximately 0.12 mm in diameter. The coil is approximately 25 mm in length, has an internal diameter of approximately 1 mm and a helix pitch of approximately 420 micrometers. The void between the successive turns of the coil **24** is therefore approximately 300 micrometers.

The coil **24** of the heating element **21** is located coaxially within the channel **23** of the support. The heating element coil **24** is thus coiled within the channel **23** of the heating element support **20**. Moreover, the axis A of the coil **24** is thus parallel to the cylindrical axis B of the mouthpiece casing **15** and the longitudinal axis C of the electronic cigarette **1**. Moreover, the device **1** is configured such that the axis A of the coil **24** is substantially parallel to airflow F through the device when a user sucks on the device. Use of the device **1** by a user is later described in more detail.

The coil **24** is the same length as the support **20**, such that the ends of the coil **24** are flush with the ends of the support **20**. The outer diameter of the helix of the coil **24** is similar to the cross-sectional width of the channel **23**. As a result, the wire of the coil **24** is in contact with the surface **28** of the channel **23** and is thereby supported, facilitating maintenance of the shape of the coil **24**. Each turn of the coil is in contact with the surface **28** of the channel **23** at a contact point **29** on each of the four walls **28** of the channel **23**. The combination of the coil **24** and the support **20** provides a heating rod **30**, as illustrated in FIGS. **5**, **6** and **7**. The heating rod **30** is later described in more detail with reference to FIGS. **5**, **6** and **7**.

The inner surface **28** of the support **20** provides a surface for liquid to wick onto the coil **24** at the points **29** of contact between the coil **24** and the channel **23** walls **28**. The inner surface **28** of the support **20** also provides surface area for exposing wicked liquid to the heat of the heating element **21**.

There exists a continuous inner cavity **31** within the electronic cigarette **1** formed by the adjacent hollow interiors' of the mouthpiece casing **15** and the battery assembly casing **8**.

In use, a user sucks on the second end **17** of the mouthpiece casing **15**. This causes a drop in the air pressure throughout the inner cavity **31** of the electronic cigarette **1**, particularly at the air outlet **4**.

The pressure drop within the inner cavity **31** is detected by the pressure sensor **13**. In response to detection of the pressure drop by the pressure sensor **13**, the controller **14** triggers the provision of power from the power cell **9** to the heating element **21** via the electrical contacts **10**, **26**. The coil of the heating element **21** therefore heats up. Once the coil **17** heats up, liquid in the vaporization cavity **19** is vaporized. In more detail, liquid on the coil **24** is vaporized, liquid on the inner surface **28** of the heating element support **20** is vaporized and liquid in the portions **22** of the support **20** which are in the immediate vicinity of the heating element **21** may be vaporized.

The pressure drop within the inner cavity **31** also causes air from outside of the electronic cigarette **1** to be drawn, along route F, through the inner cavity from the air inlet **27** to the air outlet **4**. As air is drawn along route F, it passes through the vaporization cavity **19**, picking up vaporized liquid, and the air passageway **18**. The vaporized liquid is therefore conveyed along the air passageway **18** and out of the air outlet **4** to be inhaled by the user.

As the air containing the vaporized liquid is conveyed to the air outlet **4**, some of the vapor may condense, producing a fine suspension of liquid droplets in the airflow. Moreover, movement of air through the vaporizer **7** as the user sucks on the mouthpiece **2** can lift fine droplets of liquid off of the heating element **21** and/or the heating element support **20**. The air passing out of the air outlet **4** may therefore comprise an aerosol of fine liquid droplets as well as vaporized liquid.

With reference to FIGS. **5**, **6** and **7**, due to the cross-sectional shape of the channel, gaps **35** are formed between the inner surface **28** of the heating element support **20** and the coil **24**. In more detail, where the wire of the coil **24** passes between contact points **29**, a gap **35** is provided between the wire and the area of the inner surface **28** closest to the wire due to the wire substantially maintaining its helical form. The distance between the wire and the surface **28** at each gap **35** is in the range of 10 micrometers to 500 micrometers. The gaps **35** are configured to facilitate the wicking of liquid onto the coil **24** through capillary action at the gaps **35**. The gaps **35** also provide areas in which liquid can gather prior to vaporization, and thereby provide areas for liquid to be stored prior to vaporization. The gaps **35** also expose more of the coil **24** for increased vaporization in these areas.

Many alternatives and variations are possible. For example, in embodiments, the electronic vapor provision device **1** may be configured such that the coil **24** is mounted perpendicular to a longitudinal axis C of the device. Moreover, FIGS. **8** to **15** show examples of different heating rod **30** configurations.

FIG. **8** shows another example heating element support **20**. This is similar to the example above with the exception that the internal channel **23** has a circular cross-section rather than a square one. The coil **24** fits inside the channel **23** such that the coil turns are in contact with the channel walls **28**. There is greater contact between the coil **24** and the channel walls **28** than the example above, with the entire coil **24** generally in contact with the channel walls **28** rather than contact at given points **29**.

This increase in contact area means that more liquid can be transferred to the full length of the coil rather than particular points **29**. However, since the coil **24** is generally in constant contact with the heating element support **20**, less of the coil surface area is exposed. So in use, when the coil **24** heats up, there will be less vaporization surface.

These two examples show that a balance can be achieved between the amount of liquid on the coil **24** and the amount of vaporization surface exposed. This balance is varied by changing the amount of contact between the coil **24** and the channel **23** of the heating element support **20**.

FIG. **9** shows an example where the amount of contact between the coil **24** and the channel **23** walls **28** lies between the examples shown in FIGS. **7** and **8**. In this example, the channel **23** has an octagonal cross-section rather than a circle or a square. As such, the coil **24** has coil turns which are generally in contact with the channel **23** of the heating element support **20** at 8 points **29** of contact. More gaps **35** are provided by the configuration of FIG. **9** than the configuration of FIGS. **3** to **7**. Moreover, the provided gaps **35** are smaller, leading to greater capillary action at the gaps.

When compared to the channel **23** with the square cross-section, the increased contact, greater number of gaps **35** and smaller gap sizes all facilitate increased liquid transfer onto the coil **24**. The increased exposed coil **24** surface compared to the channel **23** with the circular cross-section allows for more exposed vaporization surface for increased vaporization.

In this way it can be seen that providing a heating element support **20** with an internal channel **23** having a regular polygon cross-section can be used to modify the amount of liquid transfer and the degree of vaporization by selecting the number of polygon sides. Thus, an optimum channel **23** cross-section can be selected.

In the examples above, the heating element support **20** has a cylindrical shape and therefore the outer surface cross-sectional shape is circular. This shape is advantageous because the mouthpiece **2** section is also cylindrical so the heating element support **20** can be efficiently fitted into the mouthpiece **2** to minimize wasted space.

Other outer surface cross-sectional shapes may for example be configured as shown in FIG. **10** having a heating element support **20** with a square outer cross-sectional shape.

FIG. **11** shows a heating element support **20** comprising a first support section **36** and a second support section **37**. The heating element support **20** is generally cylindrical in shape and the first support section **36** and second support section **37** are half cylinders, with generally semi-circular cross-sections, which are joined together to form the cylindrical shape of the heating element support **20**.

The first support section **36** and second support section **37** each comprise a side channel **38**, or groove **38**, running along their respective lengths, along the middle of their otherwise flat longitudinal faces. When the first support section **36** is joined to the second support section **37** to form the heating element support **20**, their respective side channels **38** together form the heating elements support **20** internal channel **23**.

In this example, the combined side channels **28** form an internal channel **23** having a square cross-sectional shape. Thus, the side channels **28** are each rectangular in cross-section. As in the examples above, the coil **24** is situated within the heating element support **20** internal channel **23**. Having a heating element support **20** that comprises two separate parts **36**, **37** facilitates manufacture of this component. During manufacturing, the coil **24** can be fitted into the side channel **28** of the first support section **36**, and the second support section **37** can be placed on top to form the completed heating element support **20**.

Other arrangements can also be considered to aid the construction of the heating element support **20** and coil **24** combination. FIG. **12** shows an example having a generally cylindrical heating element support **20** similar to that shown in FIG. **7**. However, the internal channel **23** is comprises a side channel **38** and the coil is thus not completely enclosed. The coil **24** can therefore be easily fitted into the open side channel **23**, **38**. Because the channel **23**, **38** is open, the coil **24** has coil turns that are in contact with the channel walls **28** at three points **29** of contact rather than four.

FIG. **13** shows an example similar to that shown in FIG. **12** where the heating element support **20** of FIG. **12** is a first support section **36** and a second support section **37** is arranged such that it runs along the open channel **23**, **38**, plugging the open channel **38** and thereby closing it, and providing a combined arrangement similar to that shown in FIG. **7**. Thus the coil **24** is enclosed inside an internal combined channel **23** and the coil turns are in contact with the channel **23** at four points **29** of contact, three points **29** of contact with the first support section **36** and one point **29** of contact with the second support section **36**.

FIG. **14** shows an example similar to that shown in FIG. **12** with the exception that the heating element support **20** has an outer rectangular cross-sectional shape. The coil **24** has coil turns having three points **29** of contact with the heating element support **20** channel **23**.

FIG. **15** shows an example similar to that shown in FIG. **13** where a first support section **36** has an open side channel **38** and the coil **24** is fitted in this side channel. A second support section **37** is placed next to the first support section so that the coil **24** is enclosed between the support sections providing an arrangement similar to that shown in FIG. **10**. The coil **24** has coil turns with four points **29** of contact with the heating element support **20** channel **23**, **38**, three with the first support section **36** and one with the second support section **37**. Once the first support section **36** and the second support section **37** are joined to form the support **20**, the formed support is substantially rectangular.

The wire of the coil **24** is described above as being approximately 0.12 mm thick. However, other wire diameters are possible. For example, the diameter of the coil **24** wire may be in the range of 0.05 mm to 0.2 mm. Moreover, the coil **24** length may be different to that described above. For example, the coil **24** length may be in the range of 20 mm to 40 mm.

The internal diameter of the coil **24** may be different to that described above. For example, the internal diameter of the coil **24** may be in the range of 0.5 mm to 2 mm.

The pitch of the helical coil **24** may be different to that described above. For example, the pitch may be between 120 micrometers and 600 micrometers.

Furthermore, although the distance of the voids between turns of the coil is described above as being approximately 300, different void distances are possible. For example, the void may be between 20 micrometers and 500 micrometers.

The size of the gaps **35** may be different to that described above.

In embodiments, the support **20** may be located partially or entirely within liquid store **6**. For example, the support **20** may be located coaxially within the tube of the liquid store **6**.

An air pressure sensor **13** is described herein. In embodiments, an airflow sensor may be used to detect that a user is sucking on the device **1**.

The heating element **21** is not restricted to having a uniform coil **24**. Moreover, in embodiments the coil **24** is described as being the same length as the support **20**. However, the coil **24** may be shorter in length than the support **20** and may therefore reside entirely within the bounds of the support **20**. Alternatively, the coil **24** may be longer than the support **20**.

An electronic vapor provision device **1** comprising an electronic cigarette **1** is described herein. However, other types of electronic vapor provision device **1** are possible.

Liquid may not be wicked and/or stored by the support **20** and could instead be wicked from the liquid store **6** to the coil and/or the inner surface **28** of the support **20** by a separate wicking element. In this case, the support **20** may not be porous.

Internal support channels **23** with cross-sectional shapes other than those described could be used.

The electronic vapor provision device **1** is not restricted to the sequence of components described and other sequences could be used such as the control circuit **11** being in the tip of the device **1** or the liquid store **6** being in the body **3** rather than the mouthpiece **2**.

The electronic vapor provision device **1** of FIG. **2** is described as comprising two detachable parts, the mouthpiece **2** and the body **3**, comprising the battery assembly **5**. Alternatively, the device **1** may be configured such these parts **2**, **5** are combined into a single integrated unit. In other words, the mouthpiece **2** and the body **3** may not be detachable.

Reference herein to a vaporization cavity **19** may be replaced by reference to a vaporization region.

Although examples have been shown and described it will be appreciated by those skilled in the art that various changes and modifications might be made without departing from the scope of the invention.

In order to address various issues and advance the art, the entirety of this disclosure shows by way of illustration various embodiments in which the claimed invention(s) may be practiced and provide for superior electronic vapor provision. The advantages and features of the disclosure are of a representative sample of embodiments only, and are not exhaustive and/or exclusive. They are presented only to assist in understanding and teach the claimed features. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects of the disclosure are not to be considered limitations on the disclosure as defined by the claims or limitations on equivalents to the claims, and that other embodiments may be utilized and modifications may be made without departing from the scope and/or spirit of the disclosure. Various embodiments may suitably comprise, consist of, or consist essentially of, various combinations of the disclosed elements, components, features, parts, steps, means, etc. In addition, the disclosure includes other inventions not presently claimed, but which may be claimed in future. Any feature of any embodiment can be used independently of, or in combination with, any other feature.

The invention claimed is:

1. An electronic vapor provision device comprising a power cell and a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support,

wherein one or more gaps are provided between the heating element and the inner surface of the heating element support, wherein the heating element is in contact with the heating element support at two or more locations, wherein the heating element support is elongated in a lengthwise direction, wherein the vaporizer further comprises a vaporization region configured such that in use the vaporization region is a negative pressure region, wherein at least part of the heating element is inside the vaporization region,

wherein the heating element support is substantially cylindrical, wherein the outer cross-sectional shape of the heating element support is a circle, and

wherein the heating element support includes a support channel in which the heating element is located, wherein the cross-sectional shape of the heating element support channel is a polygon.

2. The electronic vapor provision device of claim **1**, wherein the electronic vapor provision device further comprises a mouthpiece section and the vaporizer is part of the mouthpiece section.

3. The electronic vapor provision device of claim **1**, wherein the heating element is located such that the axis of the heating element is substantially parallel to airflow though the device when a user sucks on the device.

4. The electronic vapor provision device of claim **1**, wherein the heating element is the same length as the heating element support, such that ends of the heating element are flush with ends of the heating element support.

5. The electronic vapor provision device of claim **1**, wherein the distance between the heating element and the inner surface of the heating element support at each gap is in the range of 10 micrometers to 500 micrometers.

6. The electronic vapor provision device of claim **1**, wherein the heating element is not supported on its inside.

7. The electronic vapor provision device of claim **1**, wherein the heating element support is porous.

8. The electronic vapor provision device of claim **1**, wherein the vaporization region is inside the heating element support.

9. A mouthpiece section adapted to detachably couple to a battery assembly of an electronic vapor provision device comprising a power cell, wherein the mouthpiece section comprises a vaporizer, where the vaporizer comprises a heating element and a heating element support, wherein the heating element is on the inside of the heating element support, wherein one or more gaps are provided between the heating element and the inner surface of the heating element support, wherein the heating element is in contact with the heating element support at two or more locations, wherein the heating element support is elongated in a lengthwise direction, wherein the vaporizer further comprises a vaporization region configured such that in use the vaporization region is a negative pressure region, wherein at least part of the heating element is inside the vaporization region, wherein the heating element support is substantially cylindrical, wherein the outer cross-sectional shape of the heating element support is a circle, wherein the heating element support includes a support channel in which the heating element is located, and wherein the cross-sectional shape of the heating element support channel is a polygon.

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